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DevOps Learning Roadmap — Senior Engineer Path

Phase 1: Foundations Mastery (Week 1)

Linux

- File system hierarchy, permissions, ACLs
- Process management (systemd, cgroups, namespaces)
- Shell scripting (bash, sed, awk, grep)
- Performance tools: `top`, `htop`, `vmstat`, `iostat`, `strace`, `perf`
- Kernel tuning: `/proc`, `/sys`, sysctl
- Networking: `ss`, `netstat`, `tcpdump`, `iptables`, `nftables`

Networking

- OSI model deep dive
- TCP/IP, UDP, HTTP/2, HTTP/3, gRPC
- DNS resolution, zones, records
- Load balancing algorithms
- TLS/SSL, mTLS, certificate management
- VPNs, VXLANs, SDN concepts

Git

- Internals (objects, DAG, plumbing vs porcelain)
- Branching strategies: Git Flow, trunk-based, GitHub Flow
- Rebase, cherry-pick, bisect, reflog
- Monorepo strategies

- Git hooks, signed commits

Phase 2: Containers & Orchestration (Weeks 2-3)

Docker

- Image layers, Union FS, copy-on-write
- Multi-stage builds, BuildKit
- Networking modes: bridge, host, overlay
- Volume management
- Docker Compose v3+ features
- Security: rootless, seccomp, AppArmor

Kubernetes

- Architecture: etcd, API server, scheduler, controller-manager, kubelet
- Workloads: Deployment, StatefulSet, DaemonSet, Job, CronJob
- Networking: Services (ClusterIP/NodePort/LoadBalancer), Ingress, NetworkPolicy
- Storage: PV, PVC, StorageClass, CSI
- RBAC, ServiceAccounts, PSA
- HPA, VPA, KEDA autoscaling
- Helm, Kustomize
- Advanced: operators, CRDs, admission webhooks
- Service meshes: Istio, Linkerd

Phase 3: CI/CD & GitOps (Week 4)

CI/CD Pipelines

- Jenkins: Declarative pipelines, shared libraries, agents
- GitHub Actions: workflows, reusable actions, matrix builds
- GitLab CI: runners, stages, artifacts, caching
- Tekton, CircleCI (concepts)

GitOps

- ArgoCD: app of apps, sync policies, RBAC
- Flux CD: Kustomize controller, Helm controller
- Pull vs push model
- Drift detection & reconciliation

Phase 4: Infrastructure as Code (Week 5)

Terraform

- HCL syntax, providers, modules
- State management: remote backends, locking
- Workspaces, environments
- Terraform Cloud / Enterprise
- Testing: Terratest, `terraform validate`, `tflint`
- Advanced: dynamic blocks, meta-arguments, depends_on, lifecycle

Ansible

- Playbooks, roles, inventories
- Modules, filters, lookups
- Ansible Vault
- Dynamic inventory
- AWX / Ansible Tower
- Testing: Molecule

Pulumi (Concepts)

- Infrastructure as real code (Python, TypeScript, Go)
- Comparison with Terraform

Phase 5: Cloud Platforms (Week 6, Part 1)

AWS (Primary Focus)

- **Compute:** EC2, ECS, EKS, Lambda, Fargate
- **Networking:** VPC, subnets, Route53, ELB/ALB/NLB, CloudFront, Transit Gateway, Direct Connect
- **Storage:** S3, EBS, EFS, Glacier
- **Databases:** RDS, Aurora, DynamoDB, ElastiCache, Redshift
- **Security:** IAM, SCP, GuardDuty, SecurityHub, KMS, Secrets Manager
- **Monitoring:** CloudWatch, CloudTrail, X-Ray
- **IaC:** CloudFormation, CDK
- **Cost Optimization:** Reserved instances, Savings Plans, Spot

GCP / Azure (Conceptual)

- GKE, Cloud Run, Anthos
- Azure AKS, Azure DevOps, ARM templates

Phase 6: Monitoring & Observability (Week 6, Part 2)

Pillars of Observability

- Metrics, Logs, Traces, Events

Tools

- **Prometheus:** PromQL, federation, alerting rules, recording rules
- **Grafana:** dashboards, datasources, alerting
- **ELK Stack:** Elasticsearch, Logstash, Kibana, Beats
- **Loki + Promtail** (log aggregation)
- **Jaeger / Zipkin / Tempo** (distributed tracing)
- **OpenTelemetry** (unified instrumentation standard)

Phase 7: Security & Compliance (Week 7, Part 1)

DevSecOps

- Shift-left security philosophy
- SAST tools: SonarQube, Semgrep, Bandit
- DAST tools: OWASP ZAP, Burp Suite
- SCA: Snyk, Dependabot, Trivy
- Container scanning: Trivy, Clair, Twistlock

- Secrets management: HashiCorp Vault, AWS Secrets Manager
- SBOM generation: Syft, CycloneDX
- OPA/Gatekeeper (policy as code)
- Compliance: SOC 2, PCI-DSS, ISO 27001 concepts

Phase 8: SRE Practices (Week 7, Part 2)

Site Reliability Engineering

- SLI, SLO, SLA definitions and examples
- Error budgets and burn rates
- Toil measurement and elimination
- Incident management lifecycle
- Postmortem culture (blameless)
- Chaos engineering: Chaos Monkey, LitmusChaos
- Capacity planning
- On-call practices

Phase 9: Projects + Interview Prep (Week 8)

- Build all 6 projects end-to-end
- Practice system design questions daily
- Mock interviews (behavioral + technical)
- Review common failure scenarios and how you solved them

Key Certifications to Target

Cert	Level	Priority
CKA (Certified Kubernetes Administrator)	Advanced	□□□□□
CKAD (Certified Kubernetes App Developer)	Advanced	□□□□
CKS (Certified Kubernetes Security)	Expert	□□□□
AWS Solutions Architect Professional	Advanced	□□□□□
Terraform Associate	Mid	□□□
AWS DevOps Professional	Advanced	□□□□

Salary Benchmarks (2025-2026, US Market)

Role	Range
Senior DevOps Engineer (5-8 yr)	$150K–$200K
Staff DevOps/Platform Engineer (8-12 yr)	$190K–$250K
SRE (Senior)	$175K–$230K
DevSecOps Engineer (Senior)	$160K–$220K
Principal DevOps/Infra	$220K–$300K+